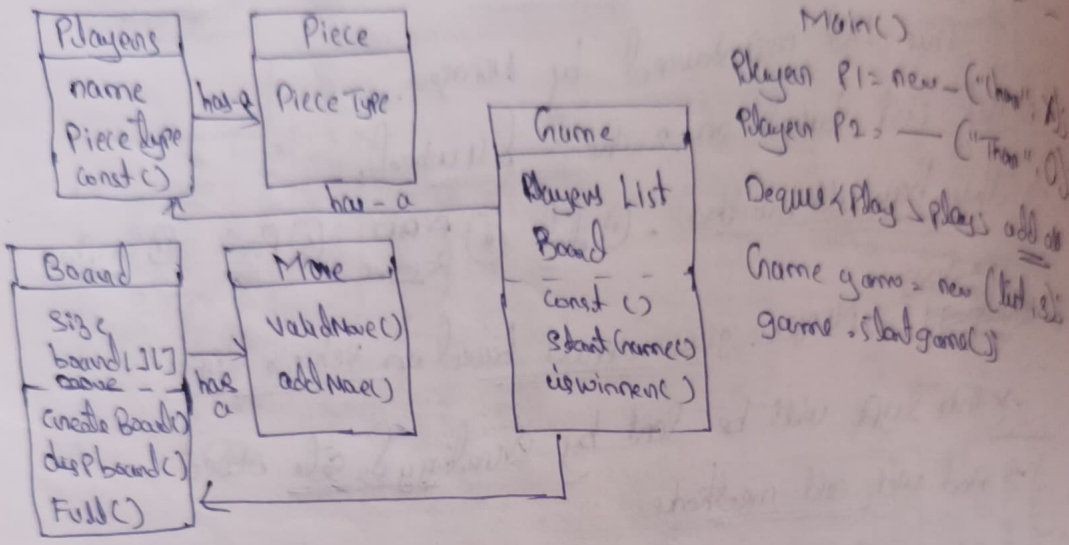
(2) TIC-TAC-TOE :- Design Game.

→ Requirements :

(1) Players. (2) Board (n×n) (3) Symbols / Pieces (4) Game Rules

→ classes / objects

(1) Players (2) Board (3) Piece (4) Game




(3) ELevator system LLD :- General Knowledge. Building, floors, were Outside Button, Lift, Inside button, display

Clarification

(1) How many Lift shall we consider.
ex :- 1, 2 .. n

(2) Lift dispatcher Algo
① → Odd/even floor
② → Fixed floors.
③ → shortest Time

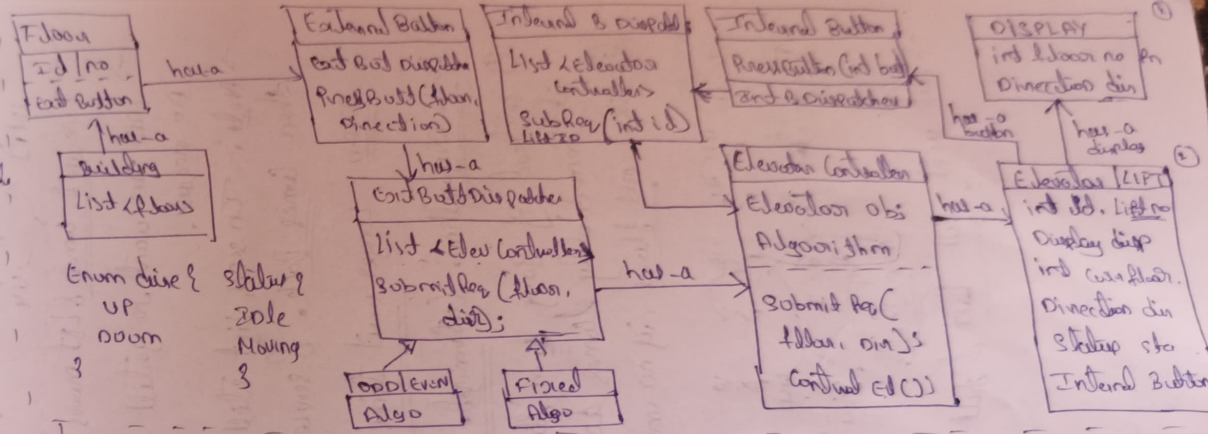
(1) served only for floors.

(2) Fixed. 1 5 6 12.

(3) shorter to user.

OBJECTS

(1) Building.
(2) Floor.
(3) External Button.
(4) Internal Button
(5) Display.
(6) Lift / Elevator Car :- Lift contains
(1) Display.
(2) Current floor.
(3) Direction — enum [UP, DOWN]
(4) status — enum [IDLE, MOVING]
(5) Internal Buttons




(1) Building will have List of all floors

(2) Each Floor has no & External Button

(3) External Button has dispatch, select direc & floor & Press

(4) Dispatcher as all List of Elevator Controller. They decide the which List should be call based on Algo.

(5) After Call Controller, they send destination floor & direction

(6) From the Internal Button, they Press the destination floor.

(7) They have dispatcher, they send floor dest & List id to the Controller.

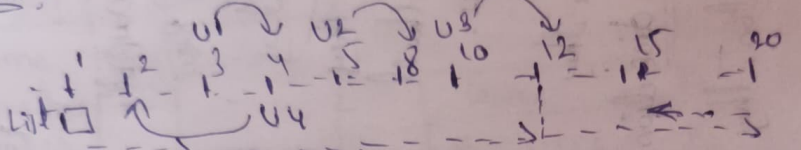
(8) Elevator Controller has some Algorithms. SCAN & Look. Based on this Algo, elevator will go up & DOWN.

(9) Inside elevator there will be display. show floor no & direction.

① SCAN ALGO:- works on moving Back & Forth sweeping motion.

① move in one direction & serves all request along the
② once reaches last, then it reverse & serves all request in opposite direction.
③ it doesn't skip floors, even there is no request.

Exam:-

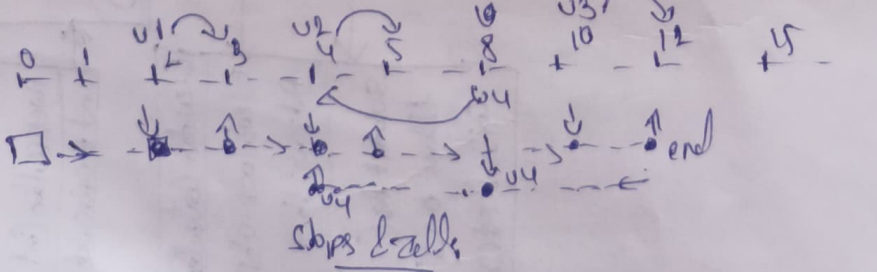



we have 20 floor Building

→ 3 required comes & List moves upto serving all 3 request
→ Last user want to go till floor 12, after that there is no request.
→ Note:- In SCAN, List still goes till last 20th floor & then comes Back. & serves all down Requests.
→ Disadv: Longer wait time. more energy usage.

② Look Algorithm:- (optimized SCAN):

① instead of going down floor 12 - 20, without any Request it immediately comes down in opp direction
② skips all floors, if no request present.




→ not went till last & till 0th floor

# Implementation & working of Look Algo.

## OBservation

① UP. Ascending order

1 2 3 4 5 8 10

-> get all UP Request & ADD to List & sort the List.

-> Then start move UP.

② Down Descending order

1 2 3 4 5 8 10

-> get all down Request. Sort them in descending & start move

---

① Instead of using List & Sorting. we use MIN & max heap.

② we get Request in accend & decend order.

---

## EDGE CASE.

① while Lift was moving on 5th floor & New Request Comes on 2nd floor.

② if we ADD to MIN heap. then we get 2. we should not go Down, until we Complete all UP Request.

③ we need to Complete it, after serving all UP & Down.

Then again while Cumming UP. we need to serve floor 2.

---

① we use PENDING JOB List. Store all Request Comes on same direction from below floor. 1=3 | 1 | 2 | 4 | Pendis.

② after serving all UP, when Comes Down, Put all pending Jobs to MIN heap.

③ same do for max heap, while Cumming Down.

---

Note: when Lift will stop. ① while going UP, if MIN HEAP becomes Empty, all UP Req Done.

② Then get all element from MAX Heap & served all Down Req.

③ Parallely, Put all pending JOBS To MIN heap